

# Harry Nguyen

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## EDUCATION

### Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

## EXPERIENCE

### TiMoto AI | Software Engineer

Sep 2025 – Present

- Architected **vLLM inference engine with PagedAttention** for production ML serving – eliminated KV cache memory fragmentation; **zero OOM failures** at production traffic with **continuous batching** achieving **sub-50ms p99**
- Architected multi-AZ ECS Fargate with **Terraform IaC, CloudWatch + Prometheus + Grafana** observability, and circuit breaker pattern – **99.9% uptime, 44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Designed and maintained **gRPC inter-service layer**; resolved production deadlock by tracing shared resource acquisition conflicts and redesigning call sequencing – **exactly-once delivery** at **sub-50ms p99**
- Led backend and infrastructure development for a 3-person engineering team – **Python (FastAPI/Django), LLM API integration** (Claude, GPT), **LangChain** orchestration, **LLM-as-a-judge evaluation pipeline** achieving **100% evaluation success rate**

### Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers – selected Protocol Buffers over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable channel serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern – **68% feature delivery acceleration** across **25K+ lines** of Chromium at **95% test coverage**; shipped to **3B+ active users**
- Collaborated with Chrome infrastructure team to deliver design documents and code reviews; changes adopted into production branch reviewed by senior Chrome engineers

### Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users, 90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with **PostgreSQL indexing** achieving **sub-100ms** for 10,000+ records

## PROJECTS

### Pulumi – Contributor | Go, TypeScript, Multi-cloud IaC

[github.com/pulumi](https://github.com/pulumi)

- Submitted **Go CLI features** enabling multi-cloud (**AWS/Azure/GCP**) K8s and infrastructure provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

## SKILLS

**ML & AI Infrastructure:** vLLM, PagedAttention, Continuous Batching, LLM APIs (Claude, GPT), LangChain, LLM-as-a-judge evaluation, PyTorch, LoRA/QLoRA

**Languages:** Python, Go, TypeScript, C++, Java, JavaScript, SQL, Bash, Rust

**Cloud & Infrastructure:** AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD, CloudWatch, Prometheus, Grafana

**Frameworks & Databases:** FastAPI, Django, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

**Distributed Systems:** gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Multi-AZ, Event-driven architecture

**AI Dev Tools:** Claude Code, GitHub Copilot, Codex, Cursor